



Morpheus: Interactive Exploration of Subspace Clustering

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ABSTRACT

Data mining techniques extract interesting patterns out of large data resources. Meaningful visualization and interactive exploration of patterns are crucial for knowledge discovery. Visualization techniques exist for traditional clustering in low dimensional spaces. In high dimensional data, clusters typically only exist in subspace projections. This subspace clustering, however, lacks interactive visualization tools. Challenges arise from typically large result sets in different subspace projections that hinder comparability, visualization and understandability.

In this work, we describe *Morpheus*, a tool that supports the knowledge discovery process through visualization and interactive exploration of subspace clusterings. Users may browse an overview of the entire subspace clustering, analyze subspace cluster characteristics in-depth and zoom into object groupings. Bracketing of different parameter settings enables users to immediately see the effects of parameters and to provide feedback to further improve the subspace clustering. Furthermore, *Morpheus* may serve as a teaching and exploration tool for the data mining community to visually assess different subspace clustering paradigms.

Categories and Subject Descriptors: H.2.8 Database management: Database applications [Data mining]

General Terms: Management

Keywords: data mining, high-dimensional data, interactive visualization tool, subspace clustering

1. INTRODUCTION

In applications such as life sciences, e-commerce, mobile information and communication, more and more data is produced that requires mining for knowledge discovery purposes. In many cases, data objects are described by many different attributes. In such high dimensional data, the so called “curse of dimensionality” poses a major challenge for data mining [5]. More specifically, clusters cannot be detected due to the increasingly similar distances between ob-

jects. As clustering aims to group similar objects while separating dissimilar ones, traditional clustering algorithms are thus hindered by irrelevant dimensions. These irrelevant dimensions differ from cluster to cluster; consequently, there are no globally irrelevant dimensions that could be removed. Recently, subspace clustering research tackles this challenge by removing locally irrelevant dimensions per cluster. Subspace clustering successfully detects patterns in subspace projections of high dimensional data. These patterns have to be intuitively presented to the user for knowledge extraction. With *Morpheus* we present a tool for intuitive representation and interactive exploration to tackle three major challenges of subspace clustering analysis:

- C1 Comparison and analysis of clusters in arbitrary subspace projections
- C2 Succinct presentation of even very large result sets for overview and in-depth studies
- C3 Visual support and feedback features for parametrization

Morpheus uses the cognitive abilities of human users to transfer the detected subspace cluster patterns to actual knowledge about the high dimensional data analyzed. *Morpheus* thus completes the final steps of the KDD (Knowledge Discovery in Databases) process for subspace clustering (cf. Figure 1).

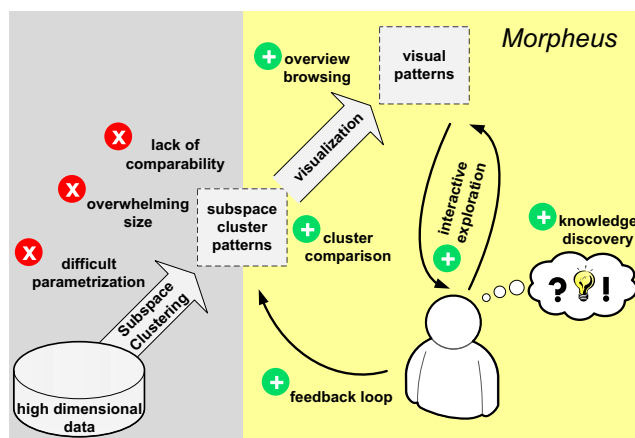


Figure 1: KDD flow in *Morpheus*

on parametrization and subsequent subspace clustering improvement. Morpheus builds on our previous work on comparison and representation in [4], providing hands on assessment of subspace clusters using the suggested measures. Additionally, it offers novel exploration features with feedback for parametrization and analysis of different subspace clustering algorithms. New 3-dimensional views allow for interactive browsing and focusing on interesting results.

2. MORPHEUS

Morpheus is a subspace clustering visualization and analysis tool that offers both overview and in-depth exploration techniques.

2.1 Visualization Techniques

Morpheus provides visualization techniques to present subspace clustering results such that users can easily gain an overview of the detected patterns, as well as an in-depth understanding of individual subspace clusters and their mutual relationship.

Gaining a meaningful overview is crucial in allowing users to assess the overall subspace clustering result. As mentioned before, subspace clustering is inherently challenging as both the typical number of resulting subspace clusters is usually enormous as well as that clusters in different projections are difficult to understand. Visualization techniques that were developed for full space clustering results rely on a common representation, i.e. no subspace projections [4]. Consequently, they cannot be applied to subspace clustering.

Our *Morpheus* system thus contains specialized techniques for visualization of subspace clustering. 2d and 3d models are an adequate representation for human cognitive abilities. Based on a recently developed comparison measure for subspace clusters the *Morpheus* system provides an overview on the entire subspace clustering result by MDS (multidimensional scaling) plots in both 2d and 3d [4]. As illustrated in Figures 2 and 3, MDS approximates distances in high dimensional spaces by two or three dimensions. While the 2d representation is a static view that allows for easy reading, the 3d MDS plots allow users to interactively explore the overall subspace clustering result. They may move around the 3d representation to focus on those subspace clusters they are most interested in. At any point, they may choose individual subspace clusters in the plot to obtain more detailed information.

Thus, MDS plots in *Morpheus* provide an overview on subspace clustering. Moreover, it helps users in interactively determining the best parameter setting. For any subspace clustering algorithm, some core parameters tend to have a large influence on the resulting output. We therefore present a bracketing representation, i.e. a series of 2d MDS plots for different parameters. Users thus get a clear visual impression of the effect of parameters and may choose the best ones for a feedback loop that generates the desired subspace clustering.

For in-depth analysis of any subspace clustering algorithm, representation of the key features of subspace clusters in a cognitively meaningful way is crucial. As subspace clustering results represent patterns in different projections by their very nature, visualization should contain information on the respective subspaces, the cluster values and additional information on the interestingness measures computed by the

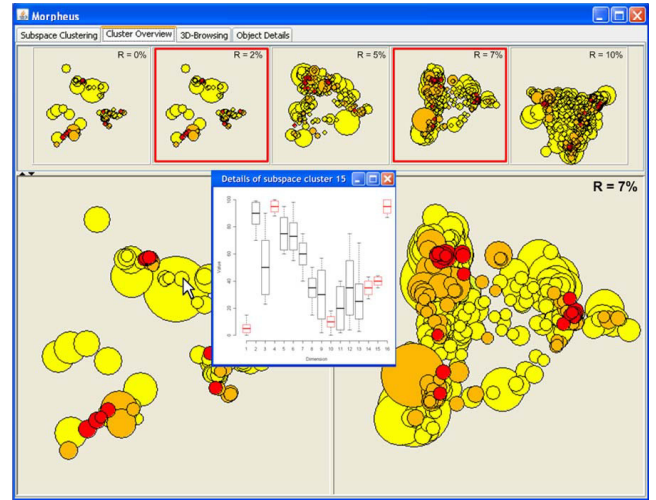


Figure 2: Screenshot of *Morpheus* application

subspace clustering algorithm. Morpheus uses a color-coding scheme where the different axis in the HSV color space are used to represent different aspects of subspace clusters in a very compact and easy to understand manner [4]. For easy navigation, subspace clusters can be zoomed into, and understood using a color legend on values of the subspace clusters.

2.2 Interactive Exploration

The conceptual design for interactive exploration in *Morpheus* is based on the *Visual Exploration Paradigm* [7]: Starting from an overview over the subspace clustering result the user can navigate through the visualized patterns. By selecting subjectively interesting subspace clusters, the user may then obtain more detailed information where desired. Detailed information is provided on three levels: for entire subspace clusterings, for single subspace clusters, as well as for individual objects. Based on the discovered knowledge, the user can give feedback to the system for further improved results. This feedback loop enables the system to use the cognitive abilities of humans for better parameter settings and thus for meaningful subspace clusters.

Overview Browsing. Interactive exploration in *Morpheus* starts from an overview of all mined subspace clusters in which the user can browse. The automatically detected patterns are thus presented to the user for a general impression of the result and a comparison of the resulting subspace clusters. As clusters are detected in arbitrary subspaces, they cannot be compared based on the full space dimensionality. *Morpheus* thus incorporates a distance function that takes the main characteristics of subspace clusters, their subspace dissimilarity and object dissimilarity into account for visualization in an MDS plot [4]. Based on such an overall distance function, subspace clusters can be intuitively represented as circles in a 2d or 3d space (Figure 2). This approximation of the original high dimensional information to a 2d or 3d representation, allows human users to easily understand the result just by the visual impression. *Morpheus* enriches this MDS information by additional visual features. The size of a subspace cluster is represented as the diameter of the circle. Its dimensionality is encoded

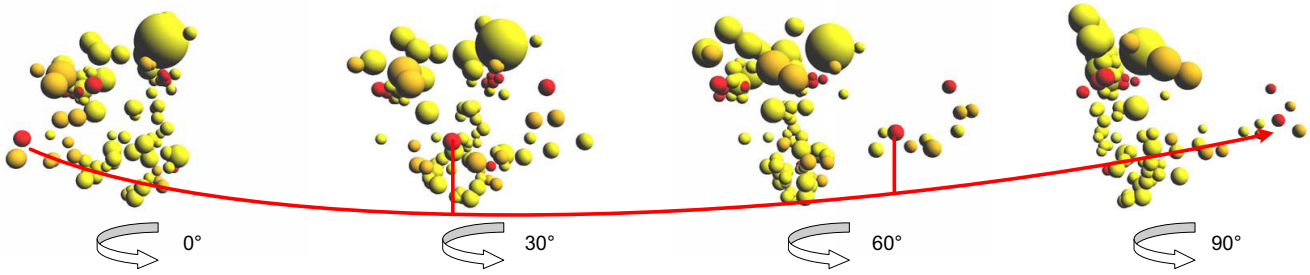


Figure 3: Detailed 3d view in *Morpheus*

by the color of the circle. This information allows users to identify similar subspace clusters, those clusters of similar dimensionality, or of similar size, or to study the overall distribution of these characteristics in the result for further analysis.

Subspace cluster details. Selecting a subspace cluster in this 2d or 3d representation by clicking on it, the user gets more detailed information which is presented in a second pop-up window. In this window, the data distribution inside the selected subspace cluster is visualized via box-plots. For each dimension the median, the 25/75% quantiles and minimum/maximum values are presented. As a characteristic information for subspace clusters, the relevant dimensions, i.e. the respective subspace projections, are highlighted in red. For the irrelevant dimensions, that are not part of the subspace projection, we may observe a scattered distribution in the values, while in the relevant dimensions the data is clustered to a tighter value range. Having this information at hand, the user can investigate the quality and internal structure of the grouping of the objects in such subspace clusters and thus discover knowledge about the underlying structure of the data.

Parameter Bracketing. Parameter setting is in general a difficult task for most data mining algorithms. This is especially the case for unsupervised techniques like clustering, where typically no prior knowledge about the data is available. This inherent problem of clustering is even more present in subspace clustering as the user has to provide parametrization for detecting clusters in different subspaces. In general the problem can be solved by guessing a parameter setting, looking at the result and then trying to choose a better parameter setting. To speed up this tedious process for users and give them more information to base their parameter choice on, *Morpheus* visualizes a set of different subspace clusters at once, called bracketing of different parameter settings for direct feedback. This means that users obtain a series of MDS plots (cf. upper part of Figure 2) from which they pick the most appropriate one(s) for subsequent runs of the subspace clustering algorithms. By directly comparing the results of different parameter settings, parametrization is no longer a guess, but becomes an informed decision based on the visual analysis of the effects of parameters. Moreover, this process is far more comfortable for users and allows reaching the desired subspace clustering result in fewer steps.

Direct subspace cluster comparison. For a more detailed analysis of two different parameter settings the user can select two clusterings out of the presented series of MDS

plots by clicking on them in the bracketing representation. These two subspace clusterings are then presented as larger plots in the lower part of the cluster overview screen in *Morpheus*. Once again, detailed information for the subspace clusters can be obtained by picking individual subspace clusters.

3d Browsing. For the overview browsing *Morpheus* provides static 2-dimensional MDS plots. These static views provide a fixed perspective for easy comparison. For in-depth browsing, where focusing to different parts of the subspace clustering is of interest, a flexible navigation through MDS plots is provided in the *Morpheus* tool. 3-dimensional MDS plot browsing allows users to shift, rotate and zoom into the MDS plot using standard 2-dimensional input devices or 3-dimensional input devices that allow for even more intuitive navigation. The user may thus identify similar or dissimilar subspace clusters that are of specific interest. In Figure 3, a 3-dimensional MDS plot is rotated in 30° steps around the z-axis.

Object Ranking. Upon selection of an interesting subspace clustering from the 2d and 3d overview browsing, the user can analyze this clustering in-depth on the object level. In the overview MDS plots, the objects are summarized by circles. For detail information on the objects and their characteristics, *Morpheus* provides an object ranking visualization technique which both provides information on the values of the objects in the cluster, as well as on the most relevant objects for the subspace clustering. The color coding, illustrated in Figure 4, uses the HSV color space, where the axes for saturation and value correspond to the interestingness measure according to the subspace clustering, i.e. interesting subspace clusters show up in bright and intensive colors in their relevant dimensions. Their value is represented as the hue axes in HSV colors. Irrelevant dimensions are denoted in black for easier reading. As subspace clusters may be large, *Morpheus* allows for zooming into areas of interest via a scroll bar to the right, which moves a zoom area, depicted via yellow lines at top and bottom. Individual objects can choose within this zoom area using the 2d or 3d input device. The user is then provided with information on the actual values of the object in question for most detailed analysis.

3. DEMONSTRATION

The demonstration setup for *Morpheus* contains the above described tools, real world data sets, precomputed examples, as well as different subspace clustering algorithms. Conference attendees may explore subspace clustering results or

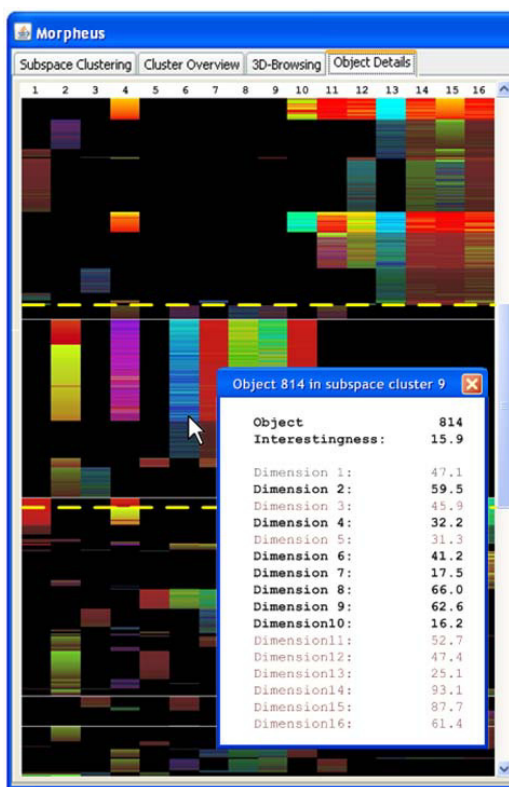


Figure 4: Object Ranking

see examples for different features and effects.

Data Sets. We use real world data from the UCI KDD archive [9]. Subspace clustering results on such real world data can be explored to find meaningful patterns in the data. Various parameter settings for the different data sets can be compared by bracketing. On meaningful subspace clusterings, the user can perform an in-depth analysis to study the structure of the respective subspace clusters. Possibly, even new or unexpected results are detected in data sets that conference attendees are familiar with.

Subspace Clustering Techniques. Besides the comparison of subspace clusterings generated on different real world data and by different parameter settings, the *Morpheus* demo includes various subspace clustering techniques from different clustering paradigms:

As a grid-based subspace clustering algorithm, *Morpheus* includes the pioneer in subspace clustering, *CLIQUE* [2]. It discretizes the data space into grid cells for efficient bottom-up processing by pruning cells which do not contain a minimum number of objects. A second grid-based algorithm in the *Morpheus* demo is *SCHISM* which extends *CLIQUE* to a variable threshold adapted to the dimensionality of the considered subspace [10].

The *Morpheus* demo includes projected clustering algorithms, that partition the objects in a search for disjoint clusters in the detected subspace projections. *PROCLUS* iteratively refines a full-space k-medoid clustering in a top-down manner on the dimensionality of the subspace projections [1]. Also incorporated in the *Morpheus* demo is a second partitioning technique, *P3C*, which combines one-dimensional cluster cores to higher-dimensional clusters bottom-up [8].

Density-based subspace clustering algorithms are available in *Morpheus* as well. *SUBCLU* mines subspace clusters in a bottom-up fashion using a monotonicity on the density assessment for pruning the search space [6]. *DUSC* normalizes the density with respect to the dimensionality of the subspace to remove dimensionality bias in monotonous density measures [3].

Extensibility and Research Impact. Further subspace clustering algorithms can be incorporated by an open interface specific for subspace clustering techniques. New algorithms, which also follow the simple input and output formats defined in the *Morpheus* system, can be easily plugged in for visualization and interactive exploration.

Morpheus may serve as a basis for teaching and research in the subspace clustering community. It allows students and researchers alike to easily see the effects of parametrization or subspace clustering paradigms. Easy extensibility makes *Morpheus* an ideal tool for the development and study of new algorithms. Consequently, *Morpheus* may even spark new interest in the domain of subspace clustering.

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